Not Running

For the best experience, choose the region closest to you. Then, start the lab to begin.

MARK UNFINISHED

Select the region closest to you

US-East

Keyboard layout: US English (US)

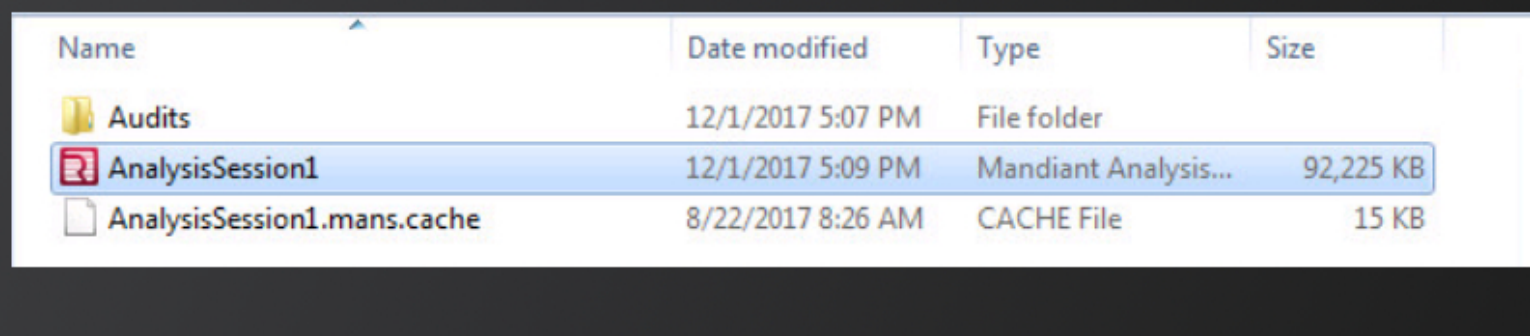
Start lab

SOLUTIONS

Below, you can find solutions for each task. Remember, though, that you can follow your own strategy, which may be different from the one explained in the following lab.

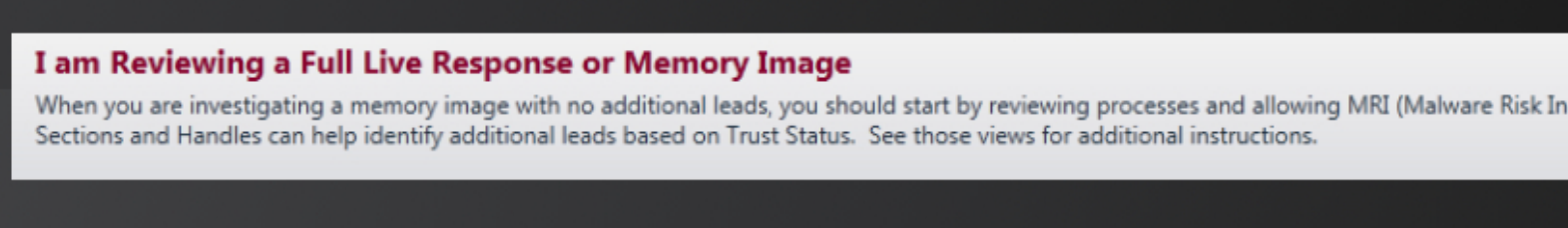
Task 1. Load saved MANS file into Redline for analysis.

1. Double-click the file within the AnalysisSession1 folder on the desktop called AnalysisSesssion1.



Name	Date modified	Type	Size
Audits	12/1/2017 5:07 PM	File folder	
AnalysisSession1	12/1/2017 5:09 PM	Mandiant Analysis...	92,225 KB
AnalysisSession1.mans.cache	8/22/2017 8:26 AM	CACHE File	15 KB

1. After Redline opens click on **I am Reviewing a Full Live Response or Memory Image** at the bottom of the Home Page or on **Processes** at the far left, in **Analysis Data**.



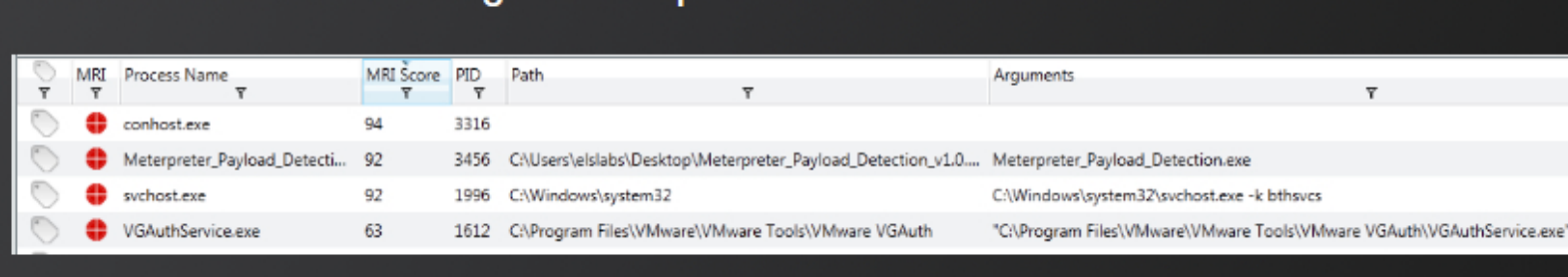
I am Reviewing a Full Live Response or Memory Image

When you are investigating a memory image with no additional leads, you should start by reviewing processes and allowing MRI (Malware Risk Index) s
Sections and Handles can help identify additional leads based on Trust Status. See those views for additional instructions.

Task 2. Analyze MANS file and answer the following questions.

1. Click on **Redlined Processes** in **Review Processes by MRI Scores** and sort the processes by MRI Score, if not already sorted.

You should see the following Redlined processes.



MRI	Process Name	MRI Score	PID	Path	Arguments
●	conhost.exe	94	3316		
●	Meterpreter_Payload_Detecti...	92	3456	C:\Users\elslabs\Desktop\Meterpreter_Payload_Detection_v1.0....	Meterpreter_Payload_Detection.exe
●	svchost.exe	92	1996	C:\Windows\system32	C:\Windows\system32\svchost.exe -k bthsvcs
●	VGAuthService.exe	63	1612	C:\Program Files\VMware\VMware Tools\VMware VGAuth	"C:\Program Files\VMware\VMware Tools\VMware VGAuth\VGAuthService.exe"

At this point we can answer the first question: "**How many processes are Redlined**?

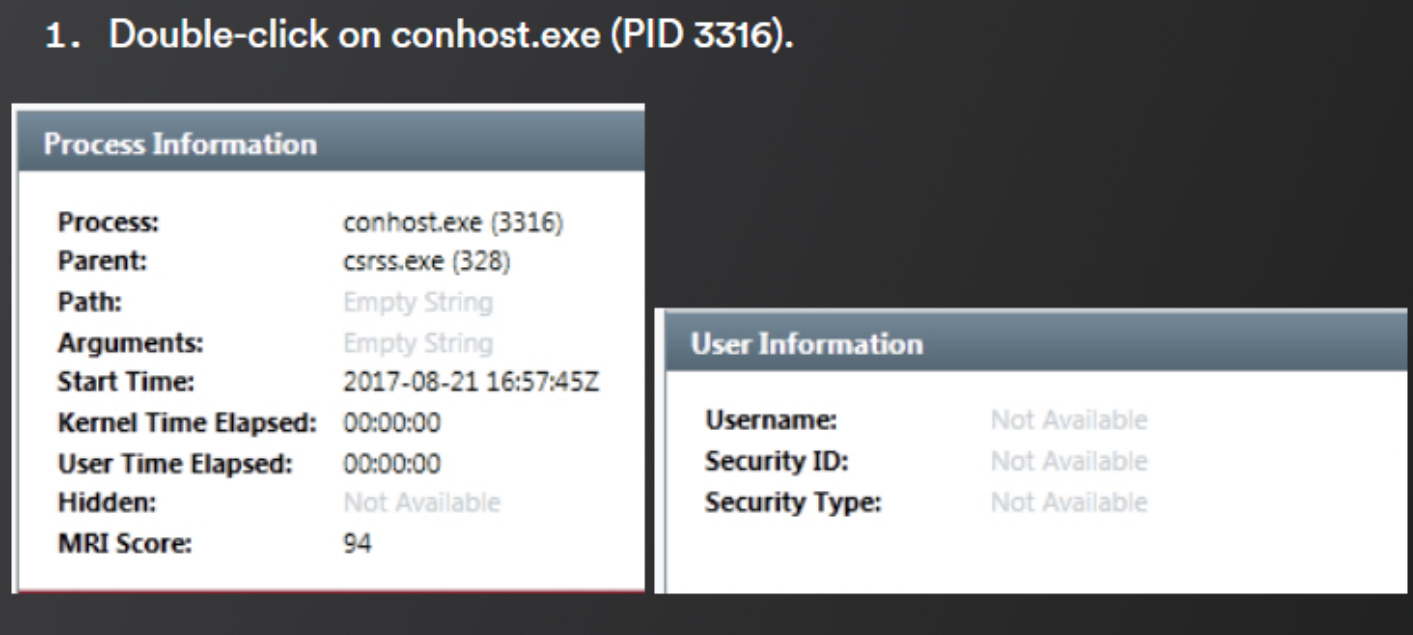
- The answer is **4**.

We can also answer the 1st part of the 2nd question: "**What are their names** and why are they Redlined?"

- The answer is **conhost.exe, Meterpreter_Payload_Detection.exe, svchost.exe, and VGAuthService.exe.**

To answer the 2nd part of the question we must look at each Redlined process individually to get more specific information about the process.

1. Double-click on conhost.exe (PID 3316).

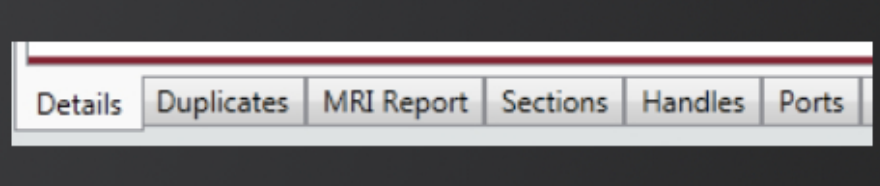


Process Information

Process: conhost.exe (3316)
Parent: csrss.exe (328)
Path: Empty String
Arguments: Empty String
Start Time: 2017-08-21 16:57:45Z
Kernel Time Elapsed: 00:00:00
User Time Elapsed: 00:00:00
Hidden: Not Available
MRI Score: 94

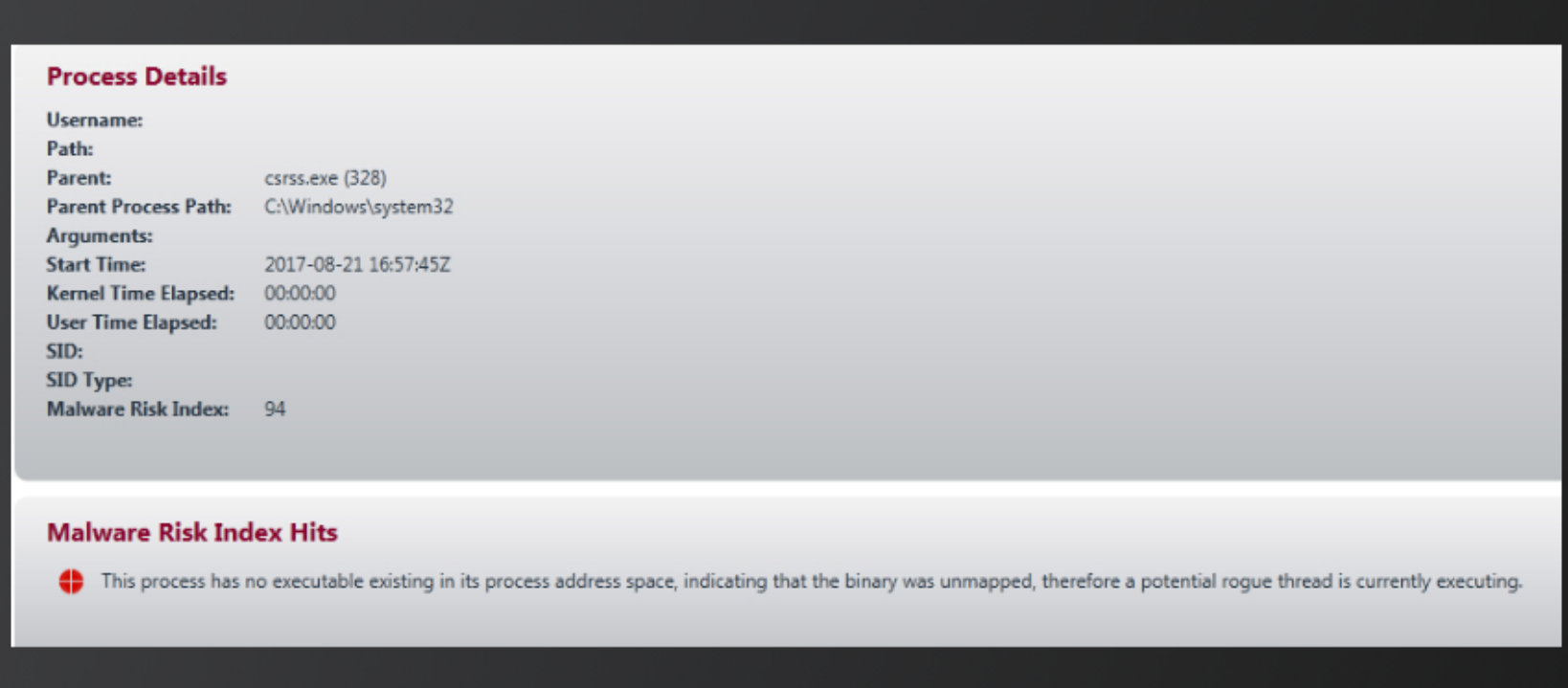
User Information

Username: Not Available
Security ID: Not Available
Security Type: Not Available



To answer the question, "**why are they Redlined?**", we must look at the MRI Report.

1. Click on MRI Report.



Process Details

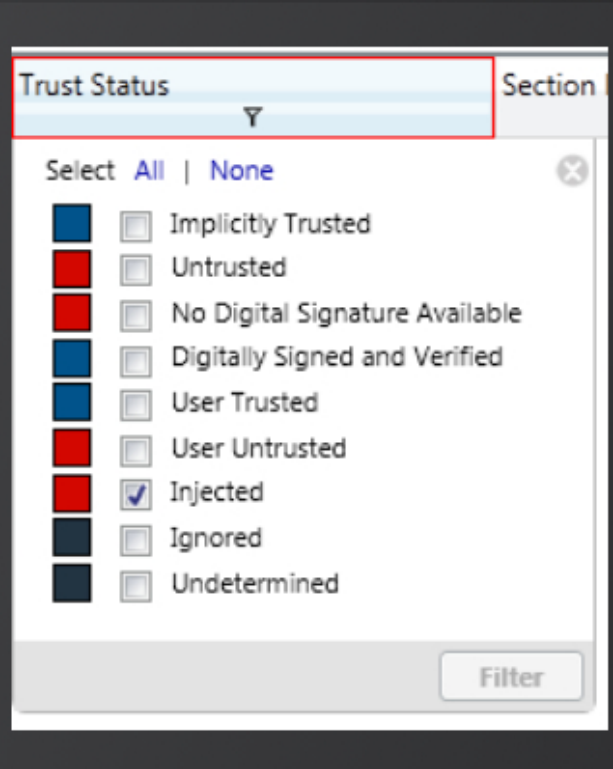
Username:
Path:
Parent: csrss.exe (328)
Parent Process Path: C:\Windows\system32
Arguments:
Start Time: 2017-08-21 16:57:45Z
Kernel Time Elapsed: 00:00:00
User Time Elapsed: 00:00:00
SID:
SID Type:
Malware Risk Index: 94

Malware Risk Index Hits

This process has no executable existing in its process address space, indicating that the binary was unmapped, therefore a potential rogue thread is currently executing.

While we have our focus on individual processes we can attempt to answer the next 2 questions by using the same tabs at the bottom.

1. Click on **Sections**, next to **MRI Report**, to see if there are any indicators of injected code.
2. Apply filter for **Injected** only and hit **Filter**.



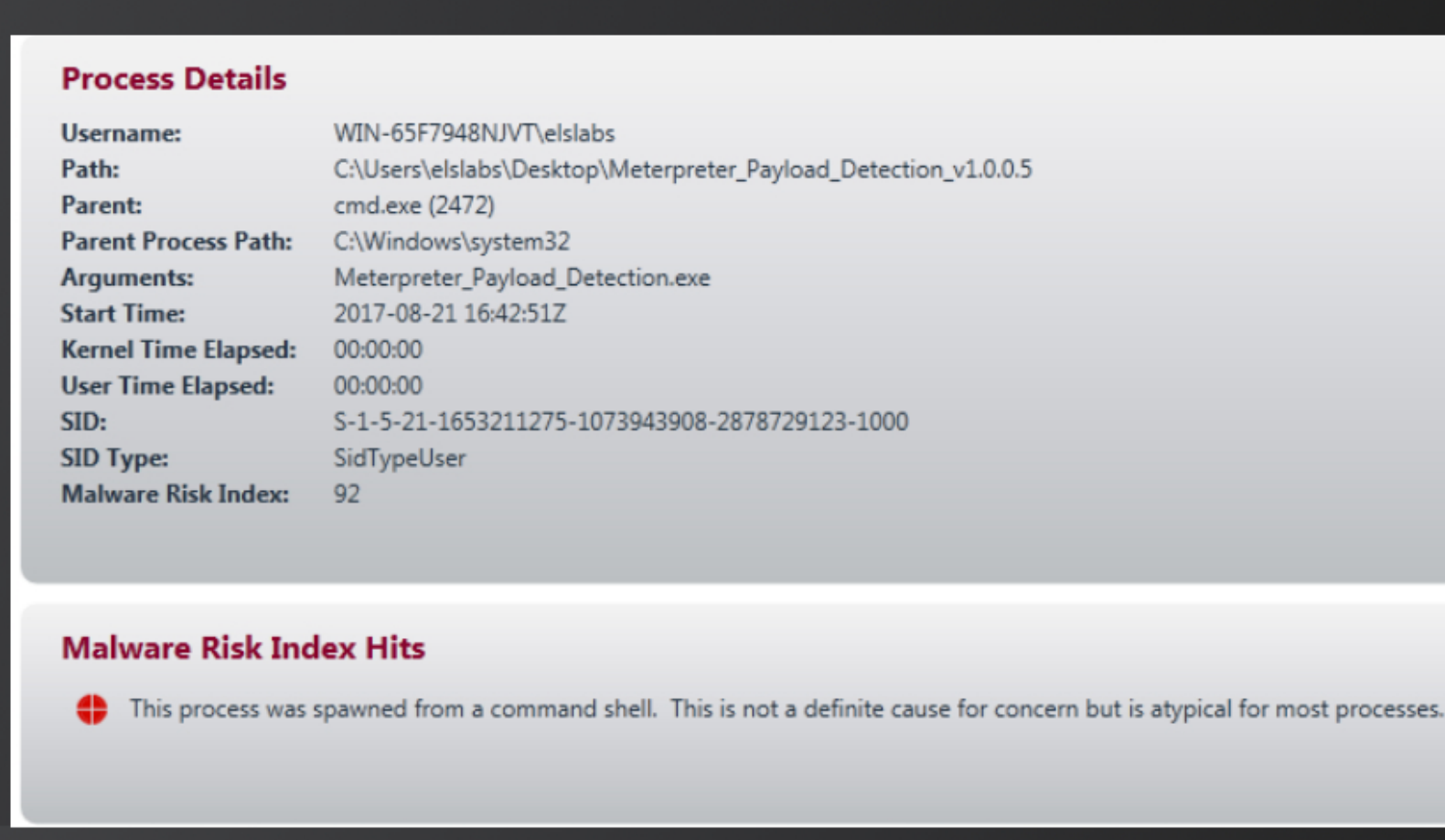
- ☐ Implicitly Trusted
- ☐ Untrusted
- ☐ No Digital Signature Available
- ☐ Digitally Signed and Verified
- ☐ User Trusted
- ☐ User Untrusted
- ☑ Injected
- ☐ Ignored
- ☐ Undetermined

Any of the Redlined processes show signs of injected code?

- No results showing as Injected.
- Click on the **Ports** tab, next to **Handles**.

Any of the Redlined processes show indicators of potentially suspicious network traffic?

- Nothing shown under **Ports**.
- We must do the same for the other 3 processes. To do so we can either click on **Processes** at the far left, in **Analysis Data**, or in the top navigational bread crumb, Home > Host > **Processes** > Full Detailed Information.
- Double-click the next process, Meterpreter_Payload_Detection.exe.



Process Details

Username: WIN-65F7948NJVT\elslabs
Path: C:\Users\elslabs\Desktop\Meterpreter_Payload_Detection_v1.0.0.5
Parent: cmd.exe (2472)
Parent Process Path: C:\Windows\system32
Arguments: Meterpreter_Payload_Detection.exe
Start Time: 2017-08-21 16:42:51Z
Kernel Time Elapsed: 00:00:00
User Time Elapsed: 00:00:00
SID: S-1-5-21-1653211275-1073943908-2878729123-1000
SID Type: SidTypeUser
Malware Risk Index: 92

Malware Risk Index Hits

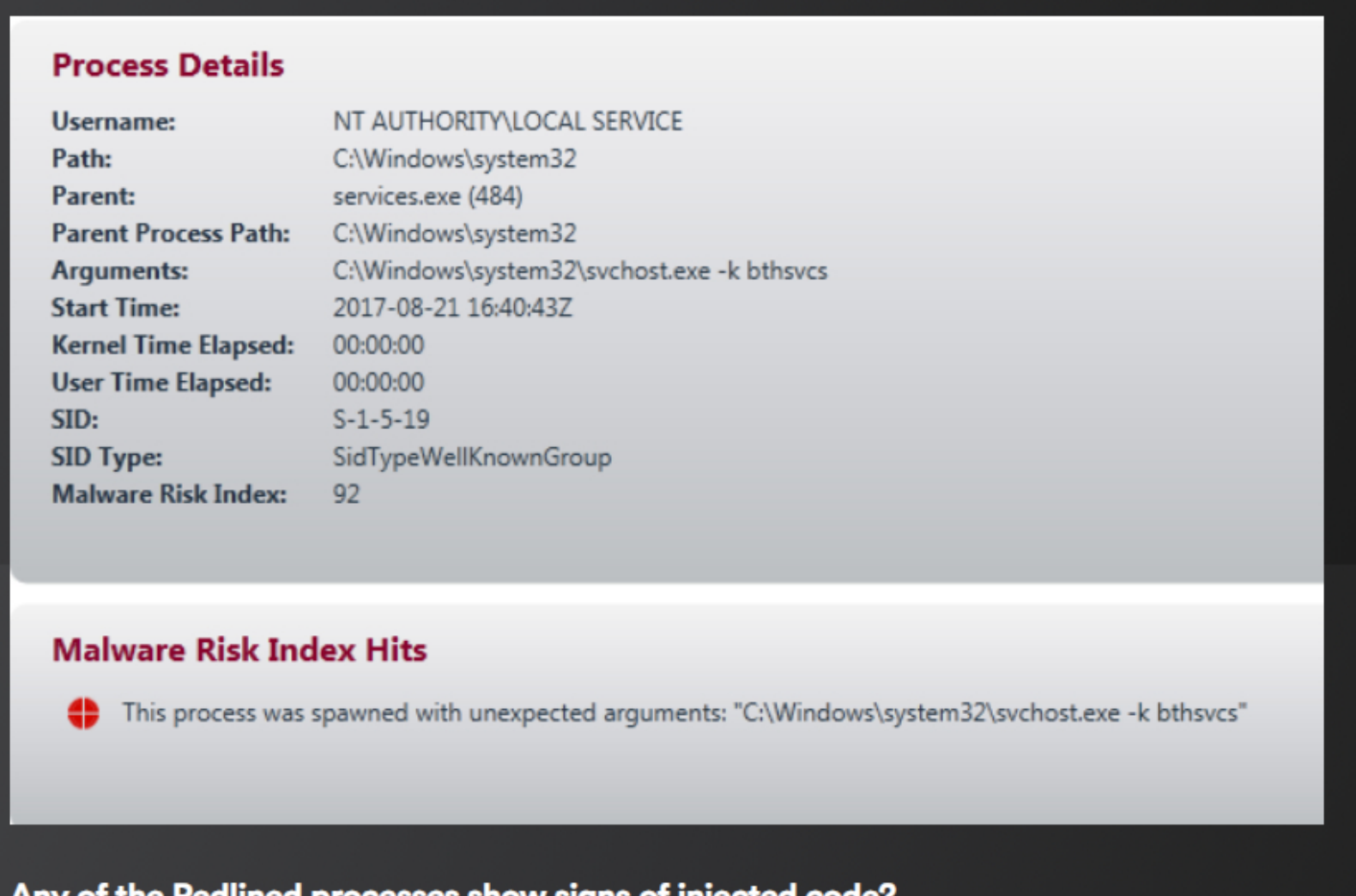
This process was spawned from a command shell. This is not a definite cause for concern but is atypical for most processes.

Any of the Redlined processes show signs of injected code?

- No results showing as Injected.

Any of the Redlined processes show indicators of potentially suspicious network traffic?

- Nothing shown under **Ports**.
- Do the same for svchost.exe and VGAuthService.exe.



Process Details

Username: NT AUTHORITY\LOCAL SERVICE
Path: C:\Windows\system32
Parent: services.exe (484)
Parent Process Path: C:\Windows\system32
Arguments: C:\Windows\system32\svchost.exe -k bthsvcs
Start Time: 2017-08-21 16:40:43Z
Kernel Time Elapsed: 00:00:00
User Time Elapsed: 00:00:00
SID: S-1-5-19
SID Type: SidTypeWellKnownGroup
Malware Risk Index: 92

Malware Risk Index Hits

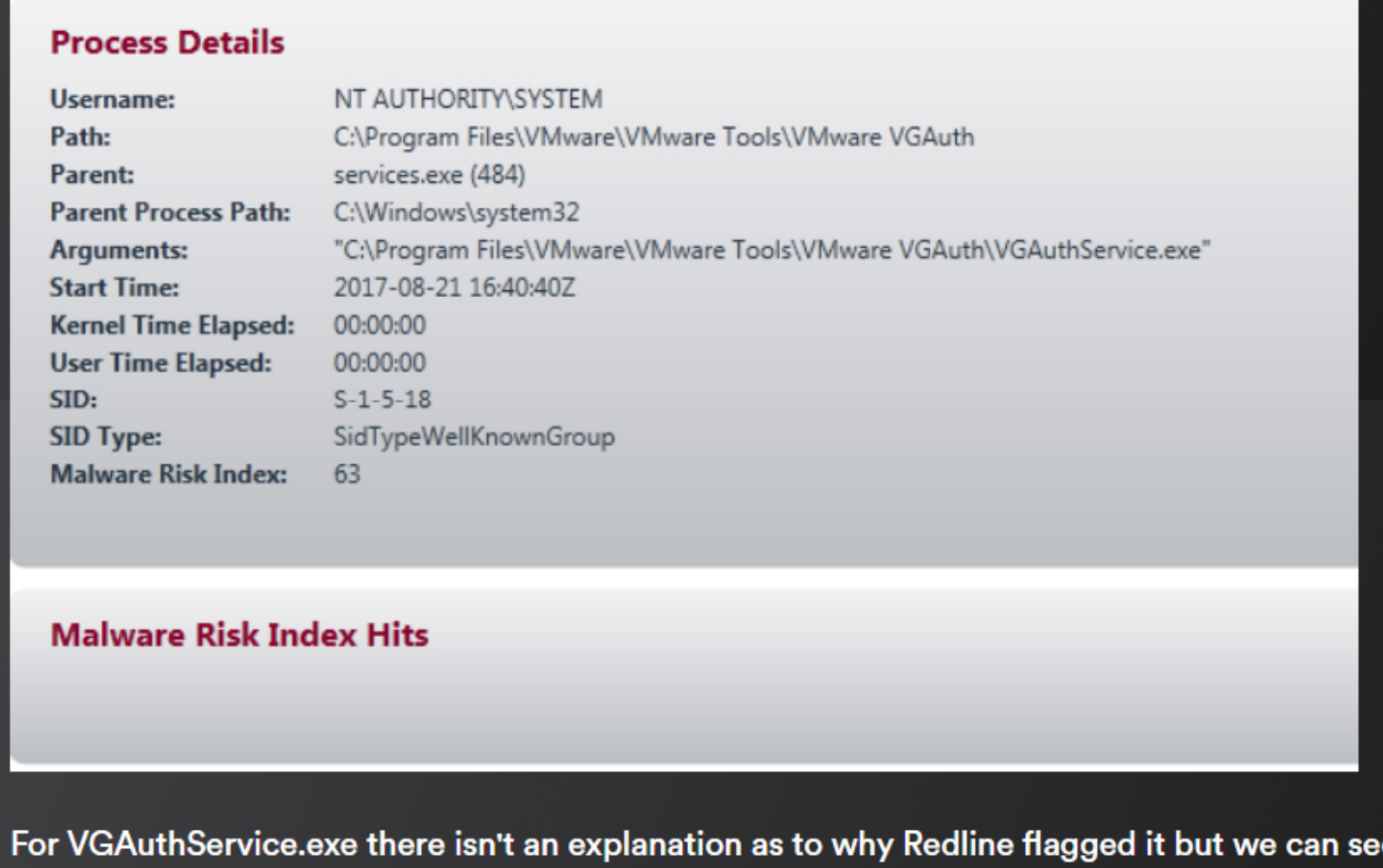
This process was spawned with unexpected arguments: "C:\Windows\system32\svchost.exe -k bthsvcs"

Any of the Redlined processes show signs of injected code?

- No results showing as Injected.

Any of the Redlined processes show indicators of potentially suspicious network traffic?

- Nothing shown under **Ports**.

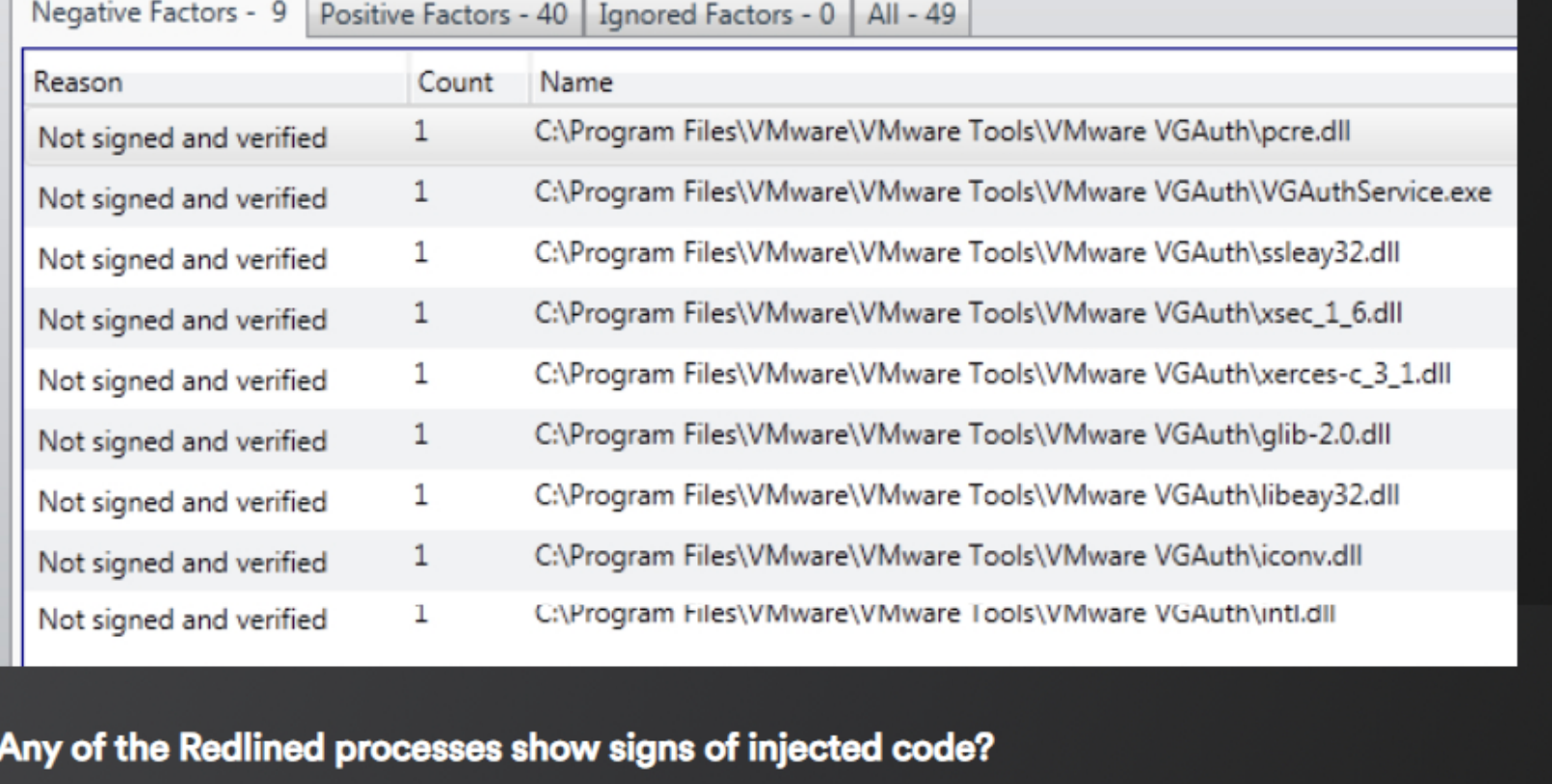


Process Details

Username: NT AUTHORITY\SYSTEM
Path: C:\Program Files\VMware\VMware Tools\VMware VGAuth
Parent: services.exe (484)
Parent Process Path: C:\Windows\system32
Arguments: "C:\Program Files\VMware\VMware Tools\VMware VGAuth\VGAuthService.exe"
Start Time: 2017-08-21 16:40:40Z
Kernel Time Elapsed: 00:00:00
User Time Elapsed: 00:00:00
SID: S-1-5-18
SID Type: SidTypeWellKnownGroup
Malware Risk Index: 63

Malware Risk Index Hits

For VGAuthService.exe there isn't an explanation as to why Redline flagged it but we can see 9 Negative Factors.



Reason	Count	Name
Not signed and verified	1	C:\Program Files\VMware\VMware Tools\VMware VGAuth\pcre.dll
Not signed and verified	1	C:\Program Files\VMware\VMware Tools\VMware VGAuth\VGAuthService.exe
Not signed and verified	1	C:\Program Files\VMware\VMware Tools\VMware VGAuth\ssleay32.dll
Not signed and verified	1	C:\Program Files\VMware\VMware Tools\VMware VGAuth\xsec_1_6.dll
Not signed and verified	1	C:\Program Files\VMware\VMware Tools\VMware VGAuth\xerces-c_3_1.dll
Not signed and verified	1	C:\Program Files\VMware\VMware Tools\VMware VGAuth\glib-2.0.dll
Not signed and verified	1	C:\Program Files\VMware\VMware Tools\VMware VGAuth\libeay32.dll
Not signed and verified	1	C:\Program Files\VMware\VMware Tools\VMware VGAuth\iconv.dll
Not signed and verified	1	C:\Program Files\VMware\VMware Tools\VMware VGAuth\intl.dll

Any of the Redlined processes show signs of injected code?

- No results showing as Injected.

Any of the Redlined processes show indicators of potentially suspicious network traffic?

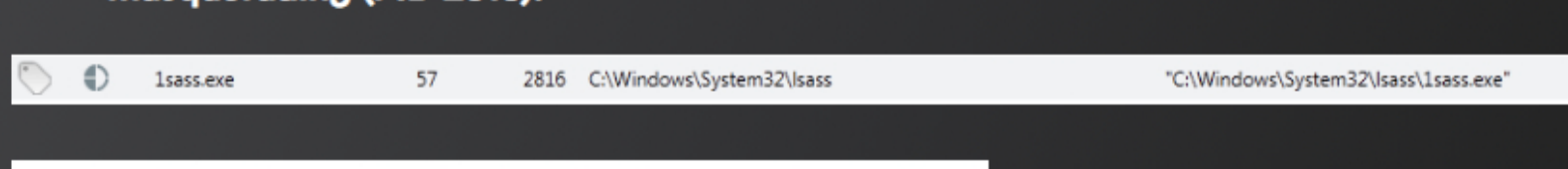
- Nothing shown under **Ports**.

Even though these 4 processes were Redlined, nothing immediately seems malicious to us. Now we must see if any processes, which weren't Redlined, are malicious. Let's look at the **Hierarchical Processes** view.

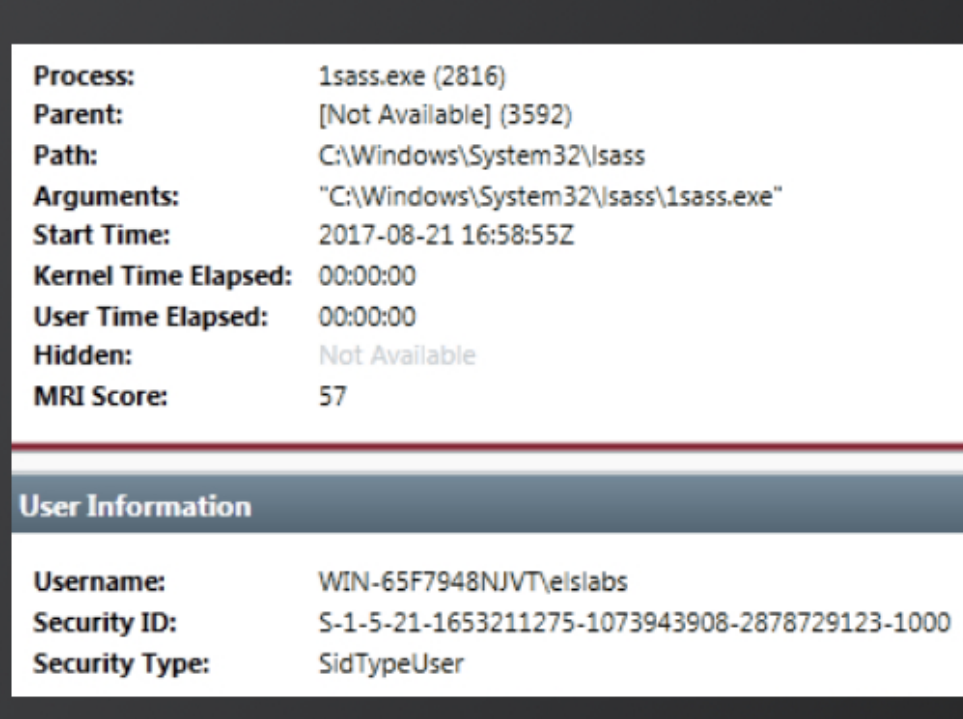
1. Click on Hierarchical Processes at the far left, in **Analysis Data**.
2. Look at all the processes, parent and child relationships, and attempt to locate the suspect process. We will not use the MRI Score.

Any other suspicious processes? If so, what are the indicators that the process is suspicious?

- **Yes. There are 2 lsass processes running: 1 is the actual process (PID 492) and 1 is masquerading (PID 2816).**



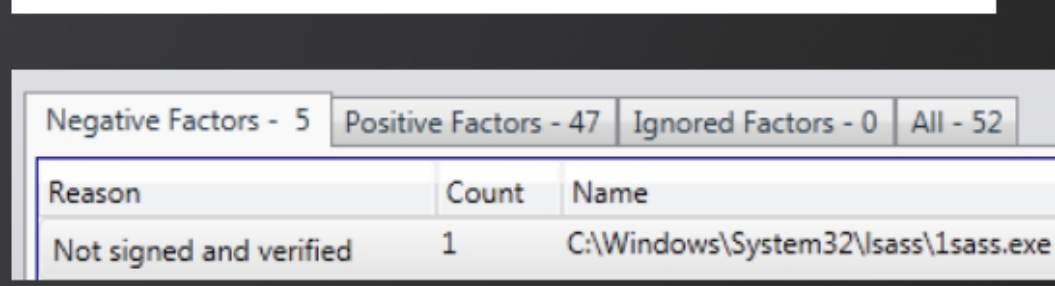
Process Name	MRI Score	PID	Path	Arguments
1sass.exe	57	2816	C:\Windows\System32\lsass	"C:\Windows\System32\lsass\1sass.exe"



Process: 1sass.exe (2816)
Parent: [Not Available] (3592)
Path: C:\Windows\System32\lsass
Arguments: "C:\Windows\System32\lsass\1sass.exe"
Start Time: 2017-08-21 16:58:55Z
Kernel Time Elapsed: 00:00:00
User Time Elapsed: 00:00:00
Hidden: Not Available
MRI Score: 57

User Information

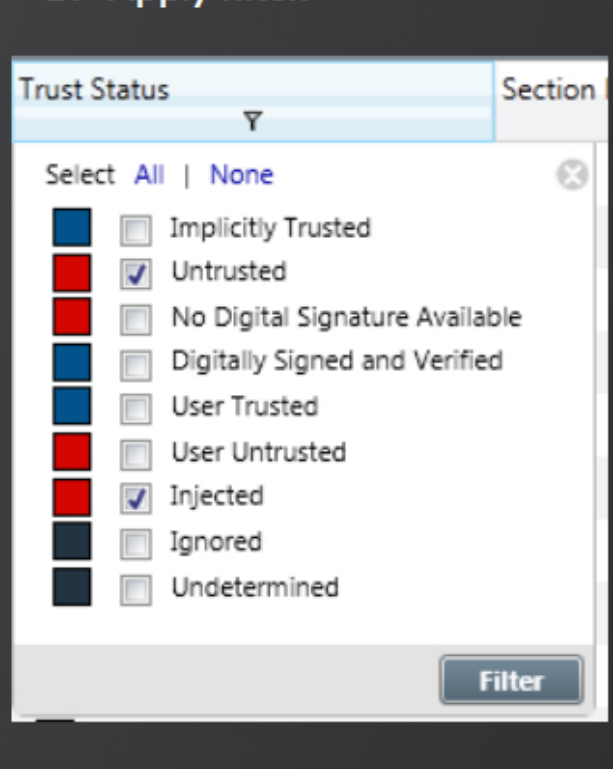
Username: WIN-65F7948NJVT\elslabs
Security ID: S-1-5-21-1653211275-1073943908-2878729123-1000
Security Type: SidTypeUser



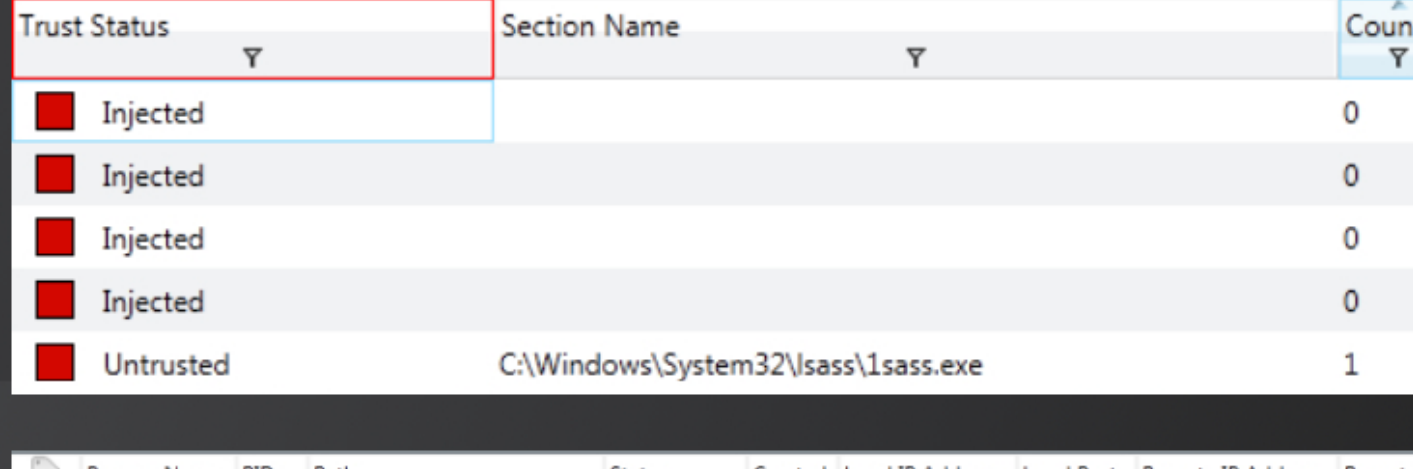
Reason	Count	Name
Not signed and verified	1	C:\Windows\System32\lsass\1sass.exe

In the **Sections** tab, towards the bottom you might already see some indicators of **Injected** and **Untrusted**. Let's filter for those only.

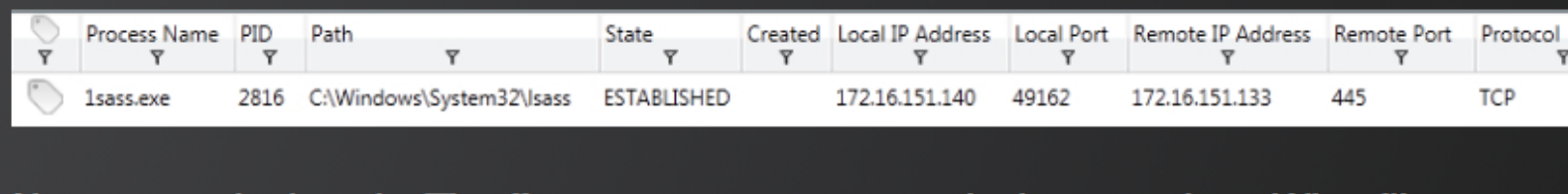
1. Apply filter.



- ☐ Implicitly Trusted
- ☑ Untrusted
- ☐ No Digital Signature Available
- ☐ Digitally Signed and Verified
- ☐ User Trusted
- ☐ User Untrusted
- ☑ Injected
- ☐ Ignored
- ☐ Undetermined



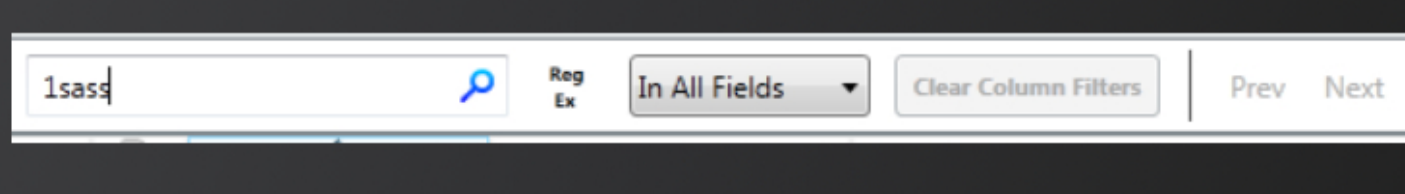
Trust Status	Section Name	Count
Injected		0
Injected		0
Injected		0
Injected		0
Untrusted	C:\Windows\System32\lsass\1sass.exe	1



Process Name	PID	Path	State	Created	Local IP Address	Local Port	Remote IP Address	Remote Port	Protocol
1sass.exe	2816	C:\Windows\System32\lsass	ESTABLISHED		172.16.151.140	49162	172.16.151.133	445	TCP

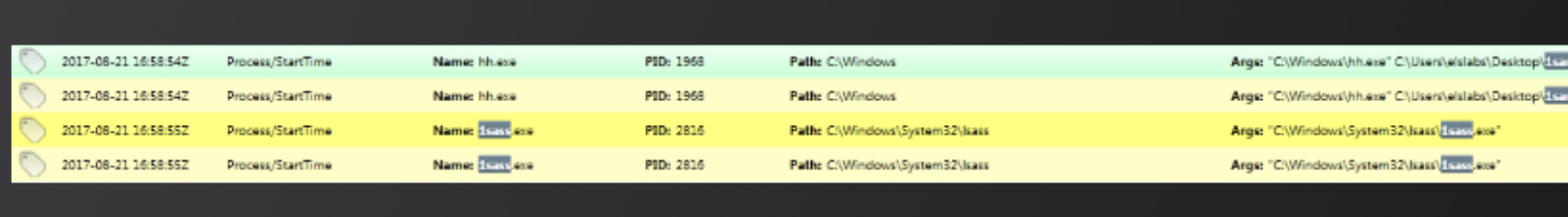
Now we can look at the **Timeline** to attempt to answer the last question, "What file most likely kicked off this suspicious binary?".

1. Click on Timeline at the far left, in **Analysis Data**.
2. Use the search box to find the term we're looking for, **1sass**.

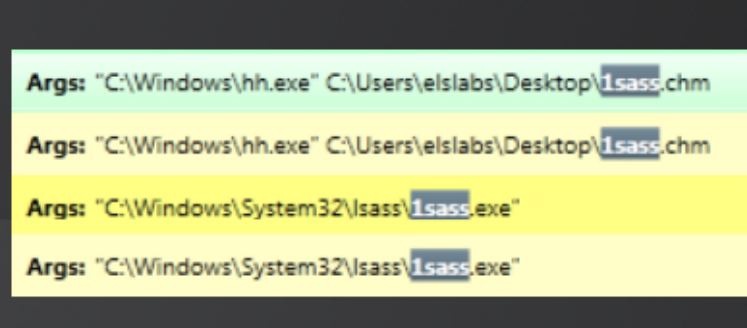


1sass

4 entries found. We can see the binary of interest, 1sass, is 2 of the 4 entries.



2017-08-21 16:58:54Z	Process/StartTime	Name: hh.exe	PID: 1968	Path: C:\Windows	Args: "C:\Windows\hh.exe" C:\Users\elslabs\Desktop\1sass.chm
2017-08-21 16:58:54Z	Process/StartTime	Name: hh.exe	PID: 1968	Path: C:\Windows	Args: "C:\Windows\hh.exe" C:\Users\elslabs\Desktop\1sass.chm
2017-08-21 16:58:55Z	Process/StartTime	Name: 1sass.exe	PID: 2816	Path: C:\Windows\System32\lsass	Args: "C:\Windows\System32\lsass\1sass.exe"
2017-08-21 16:58:55Z	Process/StartTime	Name: 1sass.exe	PID: 2816	Path: C:\Windows\System32\lsass	Args: "C:\Windows\System32\lsass\1sass.exe"



Args: "C:\Windows\hh.exe" C:\Users\elslabs\Desktop\1sass.chm
Args: "C:\Windows\hh.exe" C:\Users\elslabs\Desktop\1sass.chm
Args: "C:\Windows\System32\lsass\1sass.exe"
Args: "C:\Windows\System32\lsass\1sass.exe"

What file most likely kicked off this suspicious binary?

- **We can see a file was run on the desktop called 1sass.chm executed by hh.exe. CHM are Windows Help Files. Afterwards 1sass.exe was executed.**

Note: The information we were seeking for could have been quickly obtained by looking at **Memory Sections** at the far left, in **Analysis Data** under **Processes**. In Memory Sections, select **Injected Memory Sections** under **Review Memory Sections / DLLs**. The output would be 4 entries showing as Injected and 1sass.exe being the process.

Summary:

- How many processes are Redlined? What are they?
  - **4**
  - **conhost.exe, Meterpreter_Payload_Detection.exe, svchost.exe, VGAuthService.exe**
- Why were they Redlined?
  - **conhost.exe = This process has no executable existing in its process address space, indicating that the binary was unmapped, therefore a potential rogue thread is currently executing.**
  - **Meterpreter_Payload_Detection.exe = This process was spawned from a command shell. This is not a definite cause for concern but is atypical for most processes.**
  - **svchost.exe = This process was spawned with unexpected arguments: "C:\Windows\system32\svchost.exe -k bthsvcs"**
  - **VGAuthService.exe = No explanation shown but 9 DLLs are showing as "Not signed and verified".**
- Any of the Redlined processes show signs of injected code?
  - **Nothing visible under Sections for each process.**
- Any of the Redlined processes show indicators of potentially suspicious network traffic?
  - **Nothing visible under Ports for each process.**
- Any other suspicious processes? If so, what are the indicators that the process is suspicious?
  - **Under Memory Sections 4 entries shown as Injected. Injected column shows as checked. Process name is 1sass.exe. PID 2816.**
  - **Under Hierarchical Processes PID 2816 has a MRI score of 57.**
  - **Under Details 1sass is incorrectly spelled. It should be lsass.exe and only 1 instance should be running.**
  - **Under Details the Parent is incorrect. It should be wininit.exe.**
  - **Under Details the Path is incorrect. It should be C:\Windows\system32.**
  - **Under Details the Username should not be a user account. It should be SYSTEM.**
  - **Under MRI Report shows 5 negative factors, 1 of which is "Not signed and verified".**
  - **Under Ports 1 established connection between 172.16.151.140:49162 <> 172.16.151.133:445 (TCP)**
  - **Under Ports 1 closed connection between 172.16.151.140:49159 <> 172.16.151.133:139 (TCP)**
- What file most likely kicked off this suspicious binary?
  - **Under Timeline we see 4 entries for 1sass. hh.exe was executed right before 1sass.exe:**
  - **Name: hh.exe**
- **PID:1968**
- **"C:\Windows\hh.exe" "C:\Users\elslabs\Desktop\1sass.chm"**